Stochastic Processes

math-foundations / concept 31

1 Definitions

Definition 1 (Stochastic process)**.** *Let* $(\Omega, \mathcal{F}, P)$ *be a probability space and* $(S, \mathcal{S})$ *a measurable state space. A* stochastic process *indexed by a set* T *(typically* $T = \mathbb{N}$ *or* $T = [0, \infty)$*) is a collection of random variables*

$$X = \{X_t : t \in T\}, \qquad X_t : \Omega \to S \text{ measurable.}$$

For each fixed $\omega \in \Omega$*, the function* $t \mapsto X_t(\omega)$ *is called a* sample path.

Definition 2 (Markov property)**.** *A stochastic process* $\{X_n\}_{n\in\mathbb{N}}$ *has the* Markov property *(with respect to its natural filtration) if for every* $n \in \mathbb{N}$ *and every measurable set* $A \subseteq S$*,*

$$P\big(X_{n+1} \in A \bigm| X_0, X_1, \dots, X_n\big) \;=\; P\big(X_{n+1} \in A \bigm| X_n\big).$$

Equivalently, given the present X_n*, the future* X_{n+1} *is independent of the past* $(X_0, \dots, X_{n-1})$*.*

Definition 3 (Martingale)**.** *Let* $\{\mathcal{F}_n\}_{n\in\mathbb{N}}$ *be a filtration on* $(\Omega, \mathcal{F}, P)$*. A process* $\{X_n\}$ *is a* martingale *with respect to* $\{\mathcal{F}_n\}$ *if*

1. X_n *is* $\mathcal{F}_n$*-measurable for every* n *(adaptedness);*
2. $E[|X_n|] < \infty$ *for every* n *(integrability);*
3. $E[X_{n+1} \mid \mathcal{F}_n] = X_n$ *almost surely (fair-game property).*

2 Simple Random Walk

Theorem 1. *The simple random walk* $S_n = X_1 + \cdots + X_n$ *with iid* $X_i \in \{-1, +1\}$ *each with probability* $1/2$ *satisfies* $E[S_n] = 0$ *and* $\mathrm{Var}(S_n) = n$*.*

Proof. Each summand X_i takes values ± 1 with equal probability, so

$$E[X_i] \;=\; (+1)\cdot\tfrac{1}{2} + (-1)\cdot\tfrac{1}{2} \;=\; 0,$$

and

$$E[X_i^2] \;=\; (+1)^2\cdot\tfrac{1}{2} + (-1)^2\cdot\tfrac{1}{2} \;=\; 1.$$

Hence $\mathrm{Var}(X_i) = E[X_i^2] - (E[X_i])^2 = 1 - 0 = 1.$

By linearity of expectation,

$$E[S_n] \;=\; E\left[\sum_{i=1}^{n} X_i\right] \;=\; \sum_{i=1}^{n} E[X_i] \;=\; \sum_{i=1}^{n} 0 \;=\; 0.$$

Because the X_i are independent, all cross-covariances vanish: $\mathrm{Cov}(X_i, X_j) = 0$ for $i \neq j$. The variance-of-a-sum identity therefore reduces to a sum of variances:

$$\mathrm{Var}(S_n) \;=\; \mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) \;=\; \sum_{i=1}^{n} \mathrm{Var}(X_i) + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j) \;=\; \sum_{i=1}^{n} 1 + 0 \;=\; n.$$

This proves $E[S_n] = 0$ and $\mathrm{Var}(S_n) = n$. $\square$

Remark. The standard deviation of S_n grows like $\sqrt{n}$, which is the diffusive scaling underlying Brownian motion (concept 32) and the noise schedule of diffusion-based generative models.